CS455

Deliverable 3

CodeClash

Ridham Divyakant Solanki (230853) Peeyush Agarwal (230749)
Srujan Sachin Bhirud (231039) Aritra Kolay (230192)
Om Chandrakant Chaudhari (230715)

https://github.com/SrujanBhirud/CodeClash

5 November 2026

The team repository is the one above. This pack is branch `feature/CC-58-demo-pack`. Published `main` is not rewritten. Pull requests #40–#46 stay as they were merged, with no Jira key and no review. CC-52 is not merged. Measurements below are from 5 November 2026 on the local compose stack `codeclash-srujan`, Docker Desktop 29.8.1, and Node 22, unless a row says otherwise.

1 Test report

1.1 Unit, integration, system, adversarial

Unit tests cover scoring, the seat filter, the claim token, similarity, and the agent policy. Integration tests drive the HTTP API: authoring through the publish check, registration and waitlist promotion, freeze, rejudge, quiz, email verification before a seat, ratings, and enqueueing a hardening run. The system check is Playwright, `e2e/core.spec.ts`, three tests, run as the `e2e` job. The adversarial check is the escape suite in `judge/src/isolation.test.ts`: a fork bomb ends as RE or TLE, a host file is not read, a time limit and a memory limit are enforced, and a stdout flood is cut off.

`npm run lint`, `npm run typecheck`, `npm test`, and `npm run test:coverage` passed on the tree that includes the sandbox fix. Counts from that coverage run:

Table 1: Tests by workspace.

Workspace	Tests	Result
packages/shared	33	pass
agent	37	pass
server	81	pass
judge	22	pass

The judge run includes the language suite (Python, JavaScript, C, C++17, Java 21, and Go all AC) and the batch suite (AC, AC, WA rather than three compile errors). On this branch, `server/src/http/auth.test.ts`, `flows.test.ts`, and `domain.test.ts` were run again after the six-digit code: 49 tests, pass. A local signup with Mailpit returned `mailed: true`, and the code in the inbox verified with HTTP 200.

1.2 Coverage

The gate is 80% lines, functions, and statements in each workspace. Branches are reported and are not part of the gate.

2 Concurrency

`server/src/domain/domain.test.ts` registers 400 distinct users into a contest of 200 seats. The coverage run recorded that case at about 87 seconds. The result is exactly 200 seats and no duplicate seat.

Table 2: Coverage on 5 November 2026, after CC-53.

Workspace	Lines	Functions	Statements	Branches
shared	94.99	95.65	94.99	85.58
agent	96.24	98.50	96.24	85.42
server	92.24	95.60	92.24	73.53
judge	94.31	87.50	94.31	83.72

The write is a filtered update: a seat is inserted only while `reserved < capacity`, and a unique partial index rejects a second active seat for the same user. Two requests cannot both observe spare capacity and both commit. The same file checks that a stale judge claim token does not apply a second verdict, and that a first solve is awarded once.

The demo contest Last seat has capacity 1 and registration open. Two browsers register. One is seated and one is waitlisted. That is the same filter, shown with a room of one.

3 Security

Table 3: Five threats.

Threat	What holds it
Submitted code escapes the host	`docker run` with capabilities dropped, a read-only root, no network, a pid limit, and an unprivileged run user. `/work` is read-only for the run.
A participant calls an admin route	The role check returns 403 and the denial is audited. The client also leaves the page.
A used refresh token is replayed	The refresh chain is revoked.
Registration oversells the room	The filtered seat update and the unique partial index.
Copied solutions are treated as independent	Winnowing similarity at 0.7 when a contest ends, and from Control. A match is a review. Nothing is disqualified automatically.

Three tests: the escape suite in `judge/src/isolation.test.ts`, the 403-and-audit case in `server/src/http/auth.test.ts`, and the refresh-replay case in the same file.

4 Performance

The before/after of the archive aggregation is the 3 October run in `tests/load/results/k6.md`. After the change, archive latency was p50 30.4 ms and p95 49.1 ms, at 99.9 requests/s, with 0% failed. The leaderboard script held p95 23.5 ms at 236.6 requests/s. Before the aggregation the archive route ran one stats query per problem. After it, the archive is one aggregation.

The 5 November rerun on this laptop, while Docker, the judge, and the agent were also running, is in the same file. Throughput stayed near the target (99.6/s on `api`, 138/s on `leaderboard`) and 0% of requests failed. Latency thresholds on the `api` script were missed on this machine (archive p95 807 ms). The leaderboard script stayed inside its 1000 ms threshold (p95 248 ms). That rerun is not the aggregation comparison.

5 Failure and recovery

`nodetests/failure/run.mjsredismongo` and `nodetests/failure/run.mjsjudge` against the running stack, compose project `codeclash-srujan`. `JUDGE_RESTART=touch` so `tsx watch` brought the judge back.

PASS redis: `/health` recovers
PASS redis: the in-flight submission is judged once, judged AC
PASS mongo: `/health` recovers
PASS mongo: seat counts match active seats
PASS mongo: new submissions are judged after the restart
PASS judge: the claimed submission gets one verdict after restart, judged AC

6 Defects

Table 4: Five defects, each with a regression.

Bug	Defect	Regression
CC-53	Compiled languages failed in CI. The compile container drops every capability, so a mode-755 temp directory rejected the binary and the verdict was CE.	languages, batch
CC-54	Concurrent registration could hand out more seats than capacity.	400 against 200
CC-55	A stale judge claim could award the same solve twice.	claim token
CC-56	A used refresh token could be replayed into a new session.	auth replay
CC-57	A participant could call an admin route.	403 and audit

CC-54, CC-55, CC-56, and CC-57 were already fixed on `main` and are marked Done. CC-53 is fixed in pull request #47, merged 5 November 2026, with `check` and `e2e` green. CC-53 is the sandbox defect that made `check` red from pull request #40 onward. CC-54 and CC-55 are the concurrency defects. CC-56 and CC-57 are the security defects. The escape suite sits beside them as the failure case for a hostile program.

7 Cost

The estimate before this pack was $0 of paid infrastructure: local Docker, the Gemini free tier, and no judge fleet. The cloud estimate, if a host were required, was one `t3.micro` with 20 GB gp3 for the hours of the slot, then `aws ec2 stop-instances`. A NAT gateway, a load balancer, and RDS were excluded because they bill immediately.

Actual cost is $0. Gemini uses `gemini-3.6-flash` on the free tier. Keys are only in gitignored env. On 5 November, `aws freetier get-free-tier-usage` listed only AWS Glue. EC2 was not in the 12-month allowance, so a `t3.micro` would have billed. No instance was started. The hours of a VM are therefore zero. The commands for an account that still has the allowance are in `deploy/README.md`.

8 Timeline

Taken from `git log --first-parent` on `origin/main`. There is no fourth sprint. Deliverable 2 dissolved it. Dates are merge dates.

9 Deployment

Not a static host. The API holds Socket.IO connections, the judge is a Redis consumer that calls `docker run`, and Mongo is a replica set.

The shape, when the account has the allowance, is one Ubuntu or Amazon Linux `t3.micro`, a public subnet, a public IP, 20 GB gp3, and 1 GB of swap on the root disk. The security group allows port 22 from the presenter's address and port 80, and nothing else. Mongo, Redis, and Mailpit stay on localhost. Caddy serves `client/dist` and proxies `/api`, `/socket.io`, `/health`, and `/metrics` to `127.0.0.1:4000`. Server, judge, and agent are systemd units. The judge uses that machine's

Table 5: What landed, from the merge log.

When	What landed
10–15 Oct	Local stack, modules, CI, sandbox, sign-in, submit, exactly-once judging, audit log (#4–#13).
18–19 Oct	Roles, contests, state machine, seats, waitlist, withdraw (#14–#19).
21–22 Oct	Contest submit, scoring, live board, authoring, tests, first solve (#20–#25).
23–27 Oct	Demo problems, failure drills, escape suite, backups, quiz, publish check, practice slot (#26–#32).
1–3 Nov	Rejudge, freeze, workers, audit search, profile (#33–#37).
4 Nov	Ratings (#38, last green `check` before the language work), metrics (#39), then #40–#46 with `check` red.
5 Nov	#47, CC-53, the compile permission. Similarity, mail, and the operator guide are in that merge.

Docker socket. `python:3.12-alpine` is pulled during setup so the live submit is not a cold pull. `CLIENT_ORIGIN` is `http://<public-ip>`. Gemini keys are in the unit environment, not in git. After setup, `scripts/demo-harden.mjs` is pointed at the public URL. The instance is stopped after the slot.

That launch was not done. The signed-in account would have been billed. The ten-minute path runs on this PC, which has the same compose stack, the same seed, Mailpit, and the saved hardening files. `deploy/README.md` is the checklist for an eligible account, including the stop command.

10 Traceability

Requirement, the issue that names it, the pull request that merged it, and the test that locks it.

Table 6: Requirements through to a test.

Requirement	Jira	PR	Test
Rejudge	CC-32	#33	flows
Freeze and reveal	CC-33	#34	flows
Workers	CC-38	#35	flows
Audit search	CC-39	#36	flows
Profile edits	CC-43	#37	auth
Ratings	CC-45	#38	flows
`/metrics`	CC-48	#39	metrics route
C, C++, Java, Go	none	#40	languages
Batch judge	none	#41	batch
IOI subtasks	none	#42	scoring
CodeContests import	none	#43	codecontests
Hardening engine	CC-35 open	#44	pipeline, ai
Hardening UI	none	#45	flows
k6 and smoke	CC-40 open	#46	tests/load
Compile permission	CC-53	#47	languages, batch
Similarity	CC-37	#47	similarity
Seats, 400 vs 200	CC-54	on main	domain
Stale claim	CC-55	on main	domain
Refresh replay	CC-56	on main	auth
Participant to admin	CC-57	on main	auth
Demo mail and seed	CC-58	this branch	auth, seed

Pull requests #40–#46 were merged with no review, and `check` was red from #40 because of CC-53. Those merges are not rewritten. #47 makes `check` able to go green. The row to say out loud is CC-53, pull request #47, `judge/src/languages.test.ts`.

11 Jira

Queried on 5 November 2026, `project = CC`. 58 issues. 39 Done. 1 In Progress (CC-28, per-user submission quota). 18 To Do, of which CC-5 through CC-15 are the eleven epics still in To Do, and CC-58 is the demo-pack task for this branch. CC-53 through CC-57 are Done.

12 Demo path

`Deliverables/demo-script.md` is the ten minutes: one submit to AC, the saved hardening run, Last seat with two browsers, a Redis restart with the PASS lines as backup, the admin page refused to a participant, and the CC-53 row. The hardening script writes status, attacks, notes, and proposals. It does not write a key. On Larger of two, with the provider set, the run completed with `aiStatus` AVAILABLE, two model calls, 1089 tokens, and attacks boundary 4, random 25, max 1, shape 0, model 1. Two model proposals were held: a contestant who takes absolute values, and one who compares the inputs as strings. With `AGENT_PROVIDER` unset, the same script finished with `aiStatus` DISABLED, zero tokens, model attacks 0, and the note "No AGENT_PROVIDER is set." That file is the one to repeat live.

The AI engineering log for this deliverable is `Deliverables/AI-Engineering-Log.tex`.